

Home Tasks - 04

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Q1.

A phone call is considered an international call when the person calling is in a different country than the person receiving the call.

What percentage of phone calls are international? Round the result to 1 decimal.

Assumption:

- The `caller_id` in `phone_info` table refers to both the caller and receiver.

`phone_calls` Table:

Column Name	Type
<code>caller_id</code>	integer
<code>receiver_id</code>	integer
<code>call_time</code>	timestamp

`phone_calls` Example Input:

<code>caller_id</code>	<code>receiver_id</code>	<code>call_time</code>
1	2	2022-07-04 10:13:49
1	5	2022-08-21 23:54:56
5	1	2022-05-13 17:24:06
5	6	2022-03-18 12:11:49

`phone_info` Table:

Column Name	Type
<code>caller_id</code>	integer
<code>country_id</code>	integer
<code>network</code>	integer
<code>phone_number</code>	string

phone_info Example Input:

caller_id	country_id	network	phone_number
1	US	Verizon	+1-212-897-1964
2	US	Verizon	+1-703-346-9529
3	US	Verizon	+1-650-828-4774
4	US	Verizon	+1-415-224-6663
5	IN	Vodafone	+91 7503-907302
6	IN	Vodafone	+91 2287-664895

Example Output:

international_calls_pct
50.0

Solution:

```

SELECT
  ROUND (
    SUM (
      CASE
        WHEN P1.COUNTRY_ID <> P2.COUNTRY_ID THEN 1
        ELSE 0
      END
    ) * 100.0
    /
    (
      COUNT (*)
    )
  , 1
) AS INTERNATIONAL_CALLS_PCT
FROM PHONE_CALLS P
JOIN PHONE_INFO P1
ON P.CALLER_ID = P1.CALLER_ID
JOIN PHONE_INFO P2
ON P.RECEIVER_ID = P2.CALLER_ID;

```